

Networking Task 01: Understanding Your Network Environment

Part A: Network Information

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Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wireless-AC 9461
Physical Address. . . . . : 
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : (Preferred)
Temporary IPv6 Address. . . . . : (Preferred)
Link-local IPv6 Address . . . . . : (Preferred)
IPv4 Address. . . . . : 10.62.180.247(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 04 June 2026 10:15:51 PM
Lease Expires . . . . . : 05 June 2026 12:18:40 AM
Default Gateway . . . . . : 
10.62.180.142
DHCP Server . . . . . : 10.62.180.142
DHCPv6 IAID . . . . . : 
DHCPv6 Client DUID. . . . . : 
DNS Servers . . . . . : 10.62.180.142
NetBIOS over Tcpip. . . . . : Enabled
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Part B: Basic Networking Concepts

What is an IP Address?

An IP address is a unique address given to every device connected to a network. It helps devices identify and communicate with each other just like a house address helps a courier reach the correct location an IP address helps data reach the correct device on a network or the internet

What is a MAC Address?

A MAC address is a unique hardware identifier assigned to a network adapter by the manufacturer. Unlike an IP address which can change a mac address usually remains the same It is used to identify devices within a local network

What is a Default Gateway?

A default gateway is the device that allows a computer to communicate with networks outside its own network...In most homes the router acts as the default gateway..whenever we access a website or online service our device sends the traffic through the gateway to reach the internet

What is DNS?

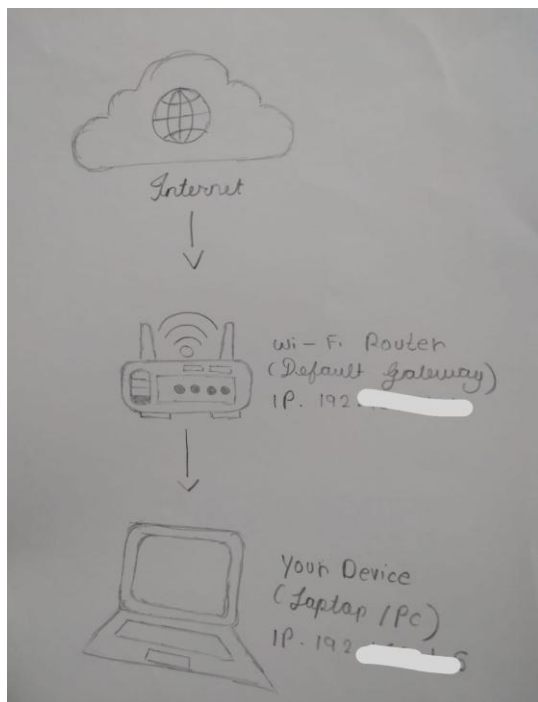
DNS stands for domain name system. It acts like the internet's phonebook by converting easy to remember website names such as google.com into numerical IP addresses that computers can understand. Without DNS users would need to remember IP addresses for every website they visit

Difference Between Public IP and Private IP

A public IP address is assigned by an internet service provider (ISP) and is used to identify a network on the internet. It is unique across the world and can be accessed from outside networks.

A private IP address is used within a local network such as a home or office network. These addresses are assigned by the router and are not directly accessible from the internet also multiple networks can use the same private IP ranges without causing conflicts.

Part C: Create a Basic Network Diagram



Part D: Network Connectivity Test

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C:\Users\ASUS>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::...
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : fe80::...
    Temporary IPv6 Address. . . . . : fe80::...
    Link-local IPv6 Address . . . . . : fe80::...
    IPv4 Address. . . . . : 10.62.180.247
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.62.180.142

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
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C:\Users\asus>ping google.com

Pinging google.com [2404:6800:4009:806::200e] with 32 bytes of data:
Reply from 2404:6800:4009:806::200e: time=263ms
Reply from 2404:6800:4009:806::200e: time=200ms
Reply from 2404:6800:4009:806::200e: time=70ms
Reply from 2404:6800:4009:806::200e: time=232ms

Ping statistics for 2404:6800:4009:806::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 70ms, Maximum = 263ms, Average = 191ms

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C:\Users\asus>tracert google.com

Tracing route to google.com [2404:6800:4009:806::200e]
over a maximum of 30 hops:
  0  1 ms    1 ms    1 ms    2401:4900:550d:8377::64
  1  146 ms   157 ms  141 ms  2401:4900:20:8::1
  2  190 ms   152 ms  91 ms   2401:4900:5a6c:1d::1
  3  85 ms    195 ms  195 ms  2404:a800:3a00:1::4dd
  4  178 ms   234 ms  159 ms  2404:a800::92
  5  222 ms   270 ms  129 ms  2404:6800:8201:340::1
  6  130 ms   *       167 ms  2404:6800:8201:340::1
  7  274 ms   287 ms  268 ms  2001:4860:0:1::55d6
  8  235 ms   188 ms  245 ms  2001:4860:0:1::17d6
  9  *        *       *       Request timed out.
 10 *        *       *       Request timed out.
 11 151 ms   281 ms  282 ms  2001:4860::9:4002:d931
 12 235 ms   157 ms  156 ms  2001:4860:0:1::1baf
 13 135 ms   160 ms  209 ms  2001:4860:0:1::7ba9
 14 158 ms   197 ms  191 ms  bom07s01-in-x0e.1e100.net [2404:6800:4009:806::200e]

Trace complete.

```

1. Was the ping successful?

Yes, the ping was successful the system received replies from google.com, which means the internet connection was working properly

2. How many hops were shown?

The traceroute showed 14 hops from my device to google.com

3. What is the purpose of traceroute?

Traceroute is a network diagnostic tool used to identify the path that data packets take from a source device to a destination server. It helps detect network delays, routing problems, and connectivity issues